

Lecture 3

IV Theory: Energy band approach

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3.1 Introduction

Most transistors operate by controlling the height of an energy barrier with an applied voltage. Examples are field-effect transistors (FET's), such as MOSFETs, JFETs (junction FET's), and HEMTs (high electron mobility transistors) as well as bipolar junction transistors (BJTs) and heterojunction bipolar transistors (HBTs) [1, 2]. The operating principles of these transistors are most readily understood in terms of energy band diagrams, a powerful approach for qualitative understanding. In the rest of these lectures, we will develop mathematical models for the *IV* characteristics, but before we do that, we need the clear physical understanding that energy band diagrams provide.

3.2 Review of energy band diagram

The operation of a semiconductor device is governed by the Poisson equation and the electron and hole continuity equations [3]. The solution to this rather complicated set of coupled, nonlinear partial differential equations gives the electrostatic potential, $\psi(\vec{r})$, and electron and hole carrier concentrations, $n(\vec{r})$ and $p(\vec{r})$, as a function of position. Drawing these energy band diagrams takes a little intuition about the solution to the semiconductor equations, but a little intuition goes a long way. Energy band diagrams are a powerful conceptual tool in the device engineer's toolkit. Essentially all of the features of transistor characteristics discussed in the previous two lectures can be understood with energy band diagrams.

An energy band diagram is a plot of the bottom of the conduction band and top of the valence band versus position. Figure 3.1 is an example that illustrates the basic principles. On the upper left is an equilibrium energy band diagram for a uniformly doped n-type semiconductor with the electron density equal to the ionized donor density, $n_0 = N_D^+$. Recall that n_0 is related to the relative position of the Fermi level with respect to the band edge according to

$$n_0 = N_C e^{(E_F - E_c)/k_B T}, \quad (3.1)$$

where N_C is the effective density-of-states of the conduction band. Mobile electrons distribute themselves to cancel the charge of the ionized donors. The result is that this semiconductor is charge-neutral.

The lower left in Fig. 3.1 illustrates what happens when we apply a bias across the semiconductor. Because the semiconductor is charge-neutral, the electric field in the semiconductor is constant. Recalling that the electric field is minus the gradient of the electrostatic potential, $\mathcal{E} = -d\psi/dx$, we see that the electric, $\mathcal{E} = -V_A/L$, points from right to left. The electric field exerts a force on electrons, $F_e = -q\mathcal{E}$ that is opposite in sign to the electric field (q is the magnitude of the charge on an electron). The force on electrons points from the left to the right in Fig. 3.1. Imagine pushing an electron from $x = L$ to $x = 0$; we need to do work against the force and push the electron “uphill,” which raises its potential energy. We conclude that the electron potential energy at $x = 0$ is higher than at $x = L$. How much lower is the electron energy at $x = L$ than at $x = 0$?

Recall that force is minus the gradient of energy, U , or

$$\frac{dU}{dx} = -F_e = q \frac{d\psi}{dx}, \quad (3.2)$$

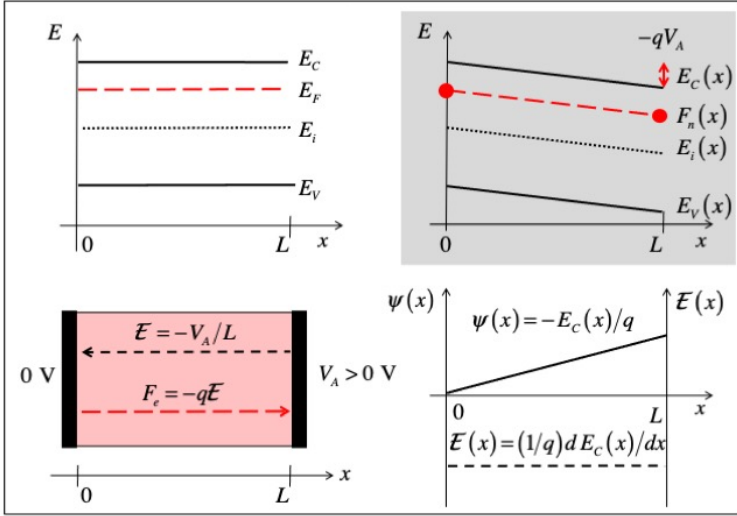


Fig. 3.1 Illustration of energy band diagram fundamentals. Upper left: Equilibrium energy band diagram of an n-type semiconductor. Lower left: Illustration of the electric field, $\mathcal{E}$, and force on electrons, F_e for the same semiconductor with an applied bias. Upper right: The energy band diagram under bias. Lower right: The electrostatic potential and electric field vs. position.

which can be integrated across the semiconductor to find

$$\int_{U(0)}^{U(L)} dU = \int_0^L -F_e dx = - \int_0^L q \left(\frac{d\psi}{dx} \right) dx \quad (3.3)$$

$$U(L) - U(0) = q(\psi(0) - \psi(L)) = q(V(0) - qV(L)) = -qV_A.$$

We use $\psi(x)$ to denote the electrostatic potential in the semiconductor and V to denote voltages on the contacts. We conclude that the applied voltage, V_A , at the right contact lowers the electron potential energy there by qV_A . Similarly, at any position, x , the electrostatic potential lowers the electron potential energy by $q\psi(x)$. The bottom of the conduction band is the potential energy of electrons in the conduction band, so

$$E_c(x) = E_{co} - q\psi(x), \quad (3.4)$$

where E_{co} is E_c where $\psi(x) = 0$. A positive electrostatic potential in the semiconductor lowers E_c . It also lowers $E_v = E_c - E_g$, where E_g is the band gap of the semiconductor.

As shown on the upper right of Fig. 3.1, we can now draw the energy band diagram of the semiconductor under bias. The equilibrium Fermi level

is replaced by the electron quasi-Fermi level, F_n , the analogous quantity out of equilibrium. (In general, there is a hole quasi-Fermi level too, but in this example, the two are identical, $F_p = F_n$.) The ideal ohmic contacts assumed connect to F_n . At $x = 0$, the potential is the same as in equilibrium, so $F_n(x = 0)$ is where the Fermi level was in equilibrium. At $x = L$, F_n is lowered by qV_A . A positive voltage applied to a contact lowers the quasi-Fermi level there.

In general, $F_n(x)$ inside the device is determined by a solution to the semiconductor equations, but we can use our intuition. Recall that the electron current is proportional to the slope of F_n , $J_n = n\mu_n \nabla F_n$ (p. 205 in [3]). Electrons flow through the semiconductor and around the external circuit providing the bias. We expect to find $n(x) = N_D^+$, just as in equilibrium, because charge neutrality will be maintained. The electric field and the electron density will be constant, so the current will be constant and, therefore, so will the slope of $F_n(x)$, as shown in the figure.

We have deduced how F_n varies with position; what about $E_c(x)$? Out of equilibrium, Eq. (3.1) becomes

$$n(x) = N_c e^{(F_n(x) - E_c(x))/k_B T}. \quad (3.5)$$

Since $n(x) = n_0$ to maintain space-charge neutrality, $E_c(x)$ must follow $F_n(x)$ to maintain the constant electron density. The result of this reasoning is the energy band diagram on the upper right.

Given an energy band diagram like that on the upper right of Fig. 3.1, we can “read” it to extract useful information. For example, by taking the gradient of $E_c(x)$, we find from Eq. (3.4),

$$\mathcal{E} = \frac{1}{q} \frac{dE_c}{dx}; \quad (3.6)$$

the electric field is proportional to the slope of the band edge, as shown on the lower right in Fig. 3.1. We also see from Eq. (3.4) that when $\psi(x)$ increases, $E_c(x)$ decreases, so we can sketch $\psi(x)$ vs. x by flipping $E_c(x)$ or $E_v(x)$ upside down, as also shown on the lower right.

With this short review of energy band diagrams, we are ready to discuss an energy band treatment of the MOSFET.

3.3 Equilibrium

A full understanding of MOSFETs requires multi-dimensional energy band diagrams, but the essentials can be conveyed in 1D. In this section, we examine $E_c(x)$ along the surface of a bulk (or planar) MOSFET as indicated in Fig 3.2. We'll show that the IV characteristics discussed in the first two lectures can be explained by understanding how $E_c(x)$ changes when voltages are applied to the contacts. Later, in Lecture 5, we will examine energy band diagrams normal to the surface, in the y -direction.

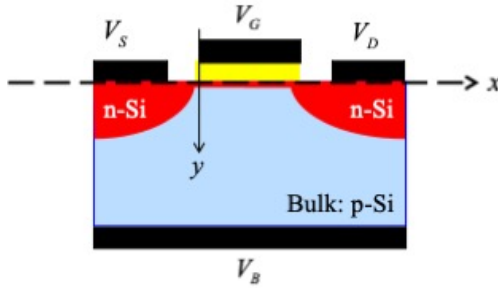


Fig. 3.2 The cross section of a bulk (or planar) MOSFET showing the line along the Si surface for which we will sketch the energy vs. position, $E_c(x, y = 0)$, from the source, across the channel, to the drain. The z -axis is into the page, in the direction of the width of the transistor, W .

The source and drain of an n-channel MOSFET are heavily doped n-type, and the channel of a bulk MOSFET like that in Fig. 3.2 is p-type. In equilibrium, the Fermi level is constant as shown in Fig. 3.3. In the n-type source and drain regions, we draw $E_c(x)$ near E_F , and in the p-type channel we draw $E_v(x)$ near E_F . After smoothly connecting the bands, we find the equilibrium energy band diagram as shown in Fig. 3.3. Note the potential energy barrier that separates electrons in the source from electrons in the drain. This barrier plays a critical role in the operation of transistors. Because the bands vary with position, we know that there is a position-dependent electrostatic potential in the semiconductor, and $E_c(x)$ varies according to Eq. (3.4). Similarly, $E_v(x) = E_c(x) - E_g$ also tracks $E_c(x)$. Because E_c is lower in the source and drain than in the channel, the electrostatic potential in the source and drain is higher than in the channel.

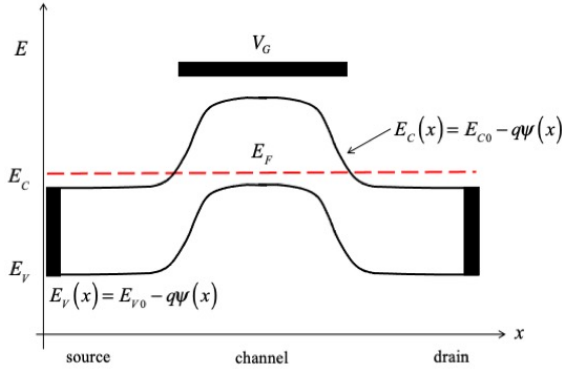


Fig. 3.3 The equilibrium energy band diagram along the top surface of an n-channel MOSFET. Also shown schematically is the gate electrode. A voltage on the gate affects the electrostatic potential in the channel. The Fermi level is above E_C in the source and drain because these regions are degenerately doped.

The next step is to understand how the energy bands change when voltages are applied to the gate and drain terminals (the source terminal is grounded). Recall that we use $\psi(x)$ to denote the electrostatic potential in the semiconductor and V to denote voltages on the contacts. When we apply a voltage, V_{DS} , between the drain and the source, the Fermi level is replaced by a position-dependent electron quasi-Fermi level, and current will flow (see [3], p. 205). When we apply a voltage, V_G , to the gate, the electrostatic potential in the semiconductor channel will be affected because the gate electrode is very close to the channel — separated only by a very thin oxide.

3.4 Application of a gate voltage

Figure 3.4 shows what happens when a positive voltage is applied to the gate. In this figure, we show only the conduction band, because we are discussing an n-channel MOSFET for which the current is carried by electrons in the conduction band. As shown on the left, for $V_S = V_D = 0$, the Fermi levels in the source, channel, and drain all align; the device is in equilibrium, and no current flows. The application of a gate voltage does not disturb equilibrium because the gate electrode is separated from the silicon by a thin gate oxide insulator. A positive gate voltage does,

however, increase the electrostatic potential in the channel, which lowers the conduction band according to Eq. (3.4)

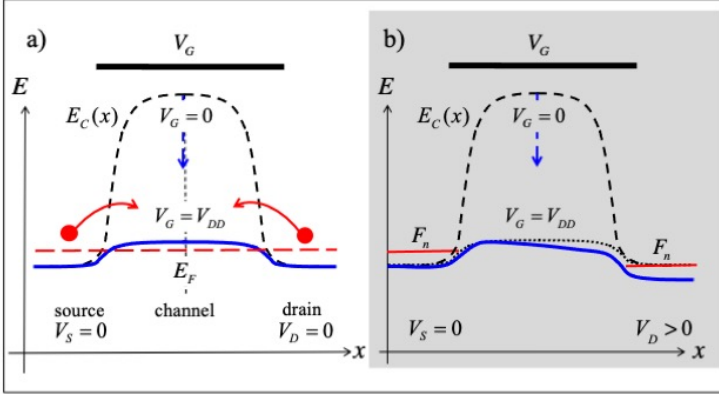


Fig. 3.4 Sketch of $E_c(x)$ for an n-channel MOSFET. Left a): The voltages on the source and drain are zero. Right b): A small positive voltage is applied to the drain. In both cases, two gate voltages are shown, a low gate voltage, $V_G = 0$, (dashed line) and a high gate voltage, $V_G = V_{DD}$, (solid line).

For low gate voltage (the dashed line for $V_G = 0$), the energy barrier between the source and drain is high, which prevents electrons from entering the channel. Under high gate voltage (the solid line for $V_G = V_{DD} > V_T$), the energy barrier is low, and the electrons' thermal energy of $k_B T$ allows electrons from the source and drain to surmount the barrier and enter the channel. As a result, when $V_G > V_T$, there is a high density of electrons in the channel. The channel, with its uniform concentration of electrons is like the n-type semiconductor in equilibrium shown on the upper left in Fig. 3.1 except that the high electron concentration is not produced by n-type doping but, rather, by the positive voltage on the gate.

It is important to note that the application of a gate voltage does not affect the Fermi level in the underlying silicon. The Fermi level in the device can only change if the source or drain voltages change, because the source and drain contacts are connected to the Fermi level in the device, and the gate electrode is not. We conclude that the application of a gate voltage simply raises or lowers the potential energy barrier between the source and drain. As shown in Fig. 3.4a, the device remains in equilibrium, and no current flows.

3.5 Application of a small drain voltage

Figure 3.4b shows what happens when we add a small drain voltage to the large gate voltage. The source is grounded, so the quasi-Fermi level in the source does not change, but the small, positive drain voltage lowers the quasi-Fermi level in the drain. Lowering F_n in the drain lowers E_c too. Electrostatics will keep the drain neutral, so $n \approx n_0 \approx N_D$, where N_D is the doping density in the drain. The quantity, $F_n - E_c$ determines the electron density, so to keep $n(x)$ at its equilibrium value, $E_c(x)$ must drop in the drain. The resulting $E_c(x)$ for a high V_G and a small $V_{DS} > 0$ is shown as the solid line in Fig. 3.4b.

For $V_{GS} > V_T$ and a small, positive V_{DS} , electrons enter the channel from the source and then flow down hill to the drain. Electrons also enter the channel from the drain, but it is harder for them to flow uphill across the channel to the source, so the net flow of electrons is from the source to the drain, which gives rise to a small drain current. As the small drain voltage increases, the net flow of electrons from the source to the drain increases. This is the linear region of the MOSFET *IV* characteristic.

Under small drain bias, the channel behaves very much like the n-type resistor shown in Fig. 3.1. We see from the slope of $E_c(x)$ in Fig. 3.4b that the electric field is constant in the channel, just as it was in the resistor of Fig. 3.1. The electron density is also constant in the channel, so $F_n(x)$ varies linearly in the channel, just as it did in the resistor of Fig. 3.1. If we were to increase the gate voltage further, there would be more electrons in the channel and its resistance would drop. Under small V_{DS} , the MOSFET behaves as a gate voltage controlled resistor.

3.6 Application of a large drain voltage

Figure 3.5 shows what happens when a large drain voltage is applied. The source is grounded, so the quasi-Fermi level in the source does not change, but the positive drain voltage lowers the quasi-Fermi level in the drain. Lowering the Fermi level lowers E_c too, to keep the drain neutral. The resulting energy band diagrams are shown in Fig. 3.5. Note that we have only shown the quasi-Fermi levels in the source and drain, but $F_n(x)$ varies smoothly across the device. Numerical simulations are needed to resolve $F_n(x)$ across the channel, but it is clear that there will be a slope to $F_n(x)$, so current will flow.

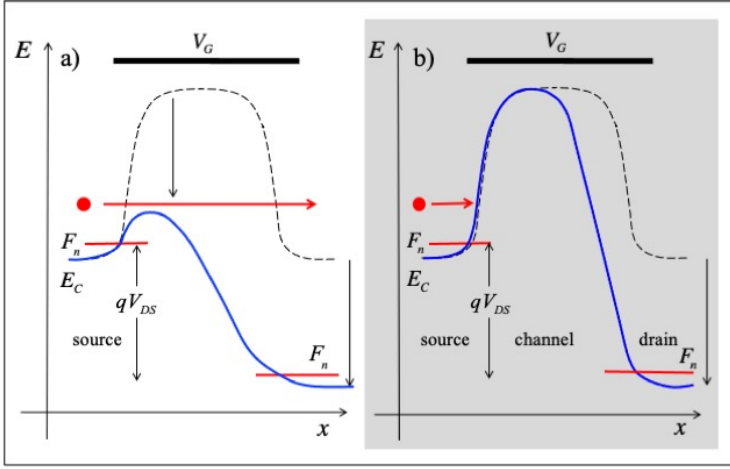


Fig. 3.5 Sketch of $E_c(x)$ for an n-channel MOSFET. The voltage on the source is zero, and the voltage on the drain is large. Left a): High gate voltage ($V_{GS} > V_T$) and high drain voltage ($V_{DS} > V_{DSAT}$). Right b): Low gate voltage ($V_{GS} = 0$) and high drain voltage. The figure on the left is for the on-state, and the figure on the right is for the off-state. In both cases, the dashed line shows the corresponding $V_{GS} = V_{DS} = 0$ equilibrium energy band diagram.

Consider first the on-state shown in Fig. 3.5a. The current is controlled by the height of the energy barrier between the source and channel. In an electrostatically well-designed MOSFET, the height of this energy barrier is strongly controlled by the gate voltage and only weakly controlled by the drain voltage. If the gate voltage increases, the barrier lowers, and more electrons in the source can surmount the barrier and flow across the channel to the drain. Electrons from the drain can also enter the channel, but they see a large energy barrier to the source, so they cannot flow across the channel to the source as some could under low drain bias. Under high drain bias, electron flow is essentially in one direction, from the source to the drain. If the drain voltage increases, it has little effect on the energy barrier at the source, so there is little effect on the drain current. This is the saturation region of the MOSFET IV characteristic. When the gate and drain voltages are high, the transistor is in the *on-state* with the current being the on-current, $I_{ON} = I_D(V_{GS} = V_{DS} = V_{DD})$ of Fig. 2.8.

Next, consider what happens when we keep the drain voltage high but reduce the gate voltage to $V_G = 0$. The resulting energy band diagram is shown in Fig. 3.5b. Note that even though V_{DS} is large, very little current

flows because there is a very small probability that electrons in the source can surmount the large source to channel energy barrier. The transistor is in the *off-state* with the current being the off-current, $I_{OFF} = I_D(V_{GS} = 0, V_{DS} = V_{DD})$ of Fig. 2.8.

3.7 Transistor operation

Figure 3.6 summarizes how modulating the height of an energy barrier with a gate voltage produces the *IV* characteristics of a MOSFET. Shown on the right are the measured output characteristics of an $L = 105$ nm n-channel MOSFET. Shown on the left are numerical simulations of $E_c(x)$ at various gate voltages in the linear and saturation regions of operation. Note that under high gate voltage in the linear region, $E_c(x)$ varies linearly with x , which corresponds to a constant electric field, as expected in the linear region where the MOSFET acts as a gate voltage controlled resistor.

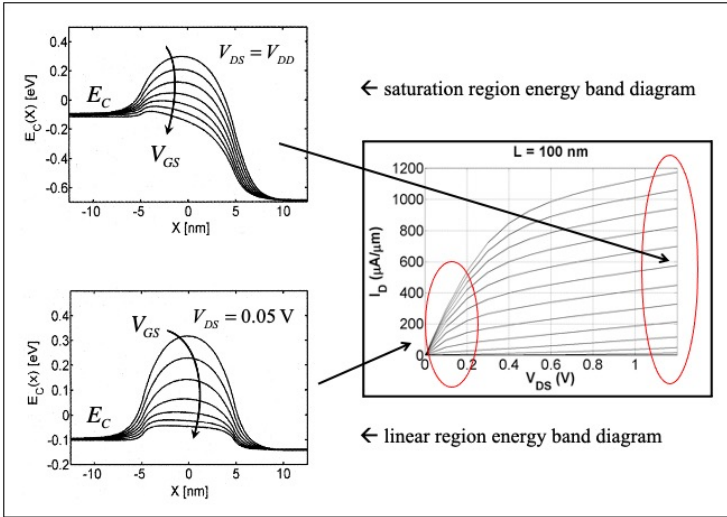


Fig. 3.6 Illustration of how the common source output characteristics of an n-channel MOSFET are explained with energy band diagrams. On the right are the measured *IV* characteristics of an $L = 105$ nm MOSFET (data provided by Shuji Ikeda, ATDF, Dec. 2007). On the left are numerical simulations of $E_c(x)$ vs. x for a different MOSFET. (©IEEE 2002. Reprinted, with permission, from: Mark Lundstrom and Zhibin Ren, “Essential Physics of Carrier Transport in Nanoscale MOSFETs,” *IEEE Trans. Electron Dev.*, **49**, pp. 133-14, 2002.)

Figure 3.6 also shows simulations of $E_c(x)$ in the saturated region. As the gate voltage pushes the potential energy barrier down, electrons in the source hop over the barrier and then flow down hill to the drain. This figure also illustrates why the drain current saturates with increasing drain voltage. It is the barrier between the source and channel that limits the current. Electrons that make it over the barrier flow down hill and out the drain. Increasing the drain voltage (assuming that it does not lower the source to channel barrier) does not increase the current. Note also that even under very high gate voltage, a small barrier remains. Without this barrier and its modulation by the gate voltage, we would not have a transistor.

Figure 3.7a shows the measured transfer characteristics of the same $L = 105$ nm MOSFET shown in Fig. 3.6. Figure 3.7b shows the measured transfer characteristic of a shorter channel length device ($L = 85$ nm) in the same technology. The DIBL observed in these characteristics is a short channel effect that will be discussed in the next section. Let's focus now on the well-behaved I_D vs. V_{GS} characteristic for $V_{DS} = 1.2$ V shown in Fig. 3.7a. We observe that below threshold, I_D varies exponentially with V_{GS} , and above threshold, I_D varies more slowly with V_{GS} because the mobile charge “screens” the channel from the influence of the gate potential. The energy band diagram can help us understand why I_D varies exponentially with V_{GS} below threshold.

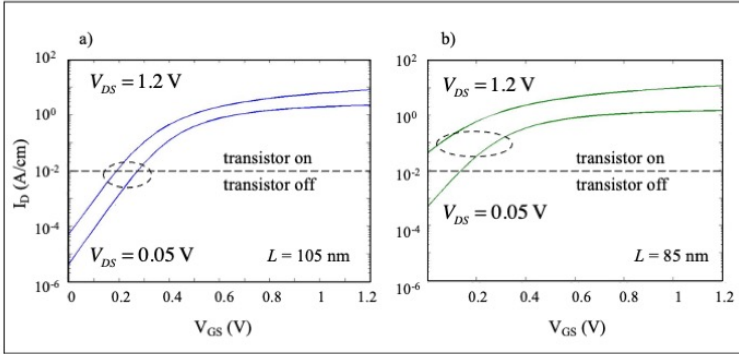


Fig. 3.7 Measured transfer characteristics of $L = 105$ nm and $L = 85$ nm MOSFETs. (Data provided by Shuji Ikeda, ATDF, Dec. 2007).

The off-current is small because the probability that an electron in the source can surmount the large off-state energy barrier shown in Fig. 3.5b is small. If E_b is the height of the energy barrier, then the probability that

an electron in the source has this energy is given by the Fermi function as

$$f(E) = \frac{1}{1 + e^{(E_b - E_F)/k_B T}} \approx e^{-(E_b - E_F)/k_B T}, \quad (3.7)$$

where the approximation comes from the fact that $E_b \gg E_F$ in the off-state. We conclude that in the off state

$$I_D \propto e^{-E_b/k_B T}. \quad (3.8)$$

Two different barrier heights produce two different currents,

$$\frac{I_{D2}}{I_{D1}} = \frac{e^{-E_{b2}/k_B T}}{e^{-E_{b1}/k_B T}} = e^{\Delta E_b/k_B T}, \quad (3.9)$$

where ΔE_b is the barrier height lowering. How much lower must E_b be to increase the current by a factor of 10? From Eq. (3.9) we find

$$\Delta E_b (eV) = (\ln 10)(k_B T/q) = 0.060 \text{ eV}. \quad (3.10)$$

The subthreshold current of a MOSFET is a result of *thermionic emission* of electrons over an energy barrier; lowering the barrier by 60 meV at $T = 300 \text{ K}$ increases the current by a factor of 10 (a decade). A positive potential in the semiconductor lowers the barrier and increases the current, so from Eq. (3.8), we find

$$I_D \propto e^{-E_c(x=0)/k_B T} \propto e^{q\psi_s(x=0)/k_B T}, \quad (3.11)$$

where $E_c(x = 0)$ is the bottom of the conduction band at the top of the barrier, and $\psi_s(x = 0)$ is the potential there. The gate voltage is what changes the potential in the semiconductor, but only a fraction of the gate voltage gets into the semiconductor. As we will see in Lecture 5,

$$\psi_s(x = 0) = \frac{V_{GS}}{m}, \quad (3.12)$$

where $m \geq 1$ is a number. Finally, using this result in Eq. (3.11), we find for the subthreshold current

$$I_D \propto e^{qV_{GS}/mk_B T}. \quad (3.13)$$

If we use Eq. (3.13) to find the change in V_{GS} needed to increase the current by a factor of 10 (i.e. the subthreshold swing), we find

$$SS = (\ln 10)m(k_B T/q) = 2.3m(k_B T/q). \quad (3.14)$$

If $m = 1$, then a change in V_{GS} produces the same change in ψ_s , and the subthreshold swing has its minimum value of 60 mV/decade at room temperature. The value of m is determined by the electrostatic design of

the MOSFET. Transistor designers work hard to make m as close to one as possible.

Figure 3.7a also shows that above threshold, I_D varies more slowly with V_{GS} . Equation (3.11) continues to apply above threshold, but Eq. (3.13) does not. As we will discuss in Lecture 5 on MOS electrostatics, above threshold it is difficult for the gate to push the energy barrier down. Instead of ψ_s being proportional to V_{GS} as in Eq. (3.12), we will find that above threshold, ψ_s is proportional to $\ln(V_{GS})$. This occurs because the high density of electrons at the top of the barrier screens out the gate potential.

3.8 Short channel effects

For many years, progress in microelectronics occurred by shrinking the channel length of MOSFETs to reduce their overall size and allow more to be placed on an integrated circuit chip. The challenge in downscaling the channel length was to deal with various *short channel effects* and preserve good electrical characteristics of the transistor. One short channel effect has to do with the output resistance, r_o , of the transistor. As shown in Fig. 3.6, I_D does not fully saturate in the so-called saturation region. The output resistance decreases as the channel length decreases. Another short channel effect is DIBL, as shown in Fig. 3.7a. As the channel length decreases, DIBL increases. Finally, it's also possible that the subthreshold swing can increase, as shown in Fig. 3.7b.

For each technology generation, there is a minimum channel length below which transistors have degraded performance. For the technology shown in Figs. 3.6 and 3.7, the $L = 105$ nm MOSFET is well behaved, but the $L = 85$ nm MOSFET is not. For the next technology generation, the transistors are re-designed to suppress short channel effects and produce transistors that are well-behaved at a shorter channel length. Although short channel effects are the result of some rather complex two- and three-dimensional electrostatics, the general principles are readily understood in terms of energy band diagrams.

Figure 3.8a illustrates the meaning of “drain-induced barrier lowering” (DIBL). In a well designed transistor, the energy barrier between the source and channel is completely controlled by the gate voltage. For very long channel MOSFETs, there is no DIBL, but for short channel MOSFETs, the positive voltage on the drain produces an electric field that can reach through to the source and lower the barrier as illustrated in Fig. 3.8a. The

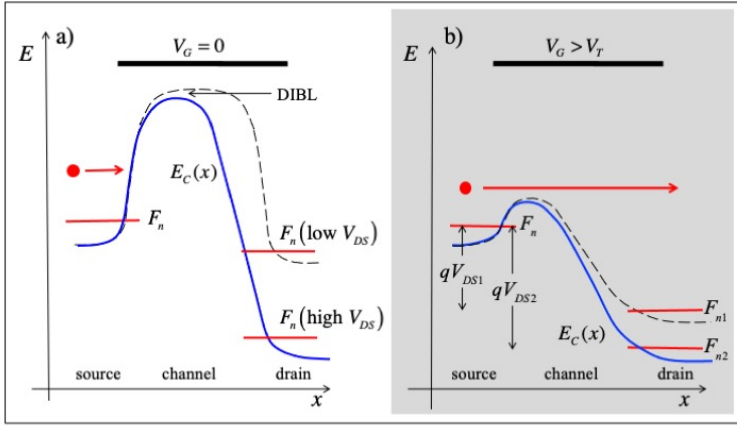


Fig. 3.8 Sketches of $E_c(x)$ that illustrate the physics of DIBL and output resistance. Left a): The off-state with low gate voltage ($V_{GS} = 0$) and two different drain voltages, a low V_{DS} (dashed line) and a high V_{DS} (solid line). Right b): The on-state with high gate voltage ($V_{GS} > V_T$) and two different drain voltages, V_{DS1} (dashed line) and $V_{DS2} > V_{DS1}$ (solid line). (Both V_{DS1} and V_{DS2} are greater than V_{DSAT} .)

lower barrier means that at $V_{GS} = 0$, the off-current is increased. Also, at any specific $V_{GS} > 0$, the current is higher because the barrier is lower. The gate voltage needed to reach a specific I_D is lower because the drain helps the gate pull the barrier down. The threshold voltage is reduced because the V_{GS} needed to push the barrier down to its on-state height is smaller. These considerations explain the IV characteristics of a well-behaved short channel transistor like the one illustrated in Fig. 3.7a.

The IV characteristic in Fig. 3.7a displays DIBL, but the subthreshold swing is essentially the same for low and high V_{DS} . For the shorter channel length MOSFET in Fig. 3.7b, SS at low V_{DS} is larger, and SS at high V_{DS} is still larger. The off-current is so high that this transistor is unusable, and it is hard to define a clear DIBL number because the two subthreshold characteristics are not parallel. We can understand what is happening with the aid of Eq. (3.12). The gate and the drain share control of the height of the energy barrier, so a change in V_{GS} does not produce the same change in ψ_s at the top of the barrier. The parameter, m , in Eq. (3.12) is greater than one, so the subthreshold swing increases according to Eq. (3.14).

Figure 3.8b illustrates a similar effect of the drain voltage for $V_{GS} > V_T$. The energy barrier is small, so the drain current is high, but if the drain voltage increases, it can lower the barrier a little, so the current increases.

This explains why the drain current does not completely saturate in a short channel transistor, i.e. why the output resistance, r_o , is not infinite.

We expect DIBL to increase, V_T to decrease, and r_o to decrease as L decreases because a reduced channel length brings the drain closer to the source, but as sketched in Fig. 3.9, if the channel length is too short, the results can be catastrophic. The transfer characteristics sketched in Fig. 3.9a show that if the channel length is too short, the gate loses control over the barrier. Figure 3.9b shows what happens when the drain has significant control of the barrier height; there is no clear saturation region, and an output resistance cannot be defined. This device is said to be *punched through* — the drain potential has reached across the channel and pulled the barrier down so that current can flow with little help from the gate. We no longer have a good transistor.

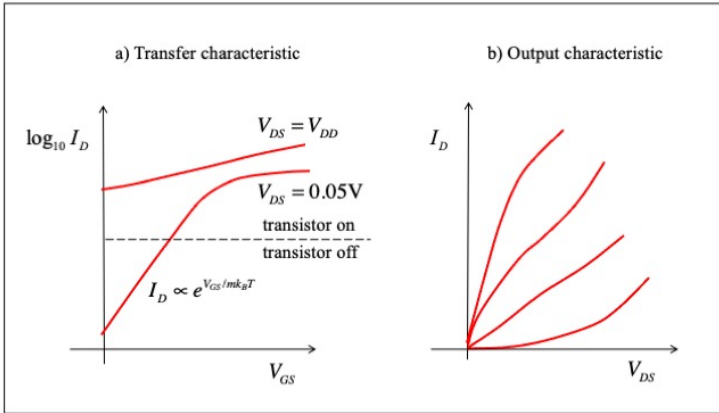


Fig. 3.9 Illustration of a MOSFET suffering from severe short channel effects. Left a): Transfer characteristics. Right b): Output characteristics.

3.9 Discussion

We have discussed how a transistor's IV characteristics can be qualitatively understood with energy band diagrams, and we have also used energy band diagrams to explain important short channel effects that occur as L is scaled down. The electrostatic design of a MOSFET to minimize short channel effects and maximize on-current at low voltage is the most

challenging aspect of MOSFET design for digital electronics. Figure 3.10 illustrates the electrostatic challenge.

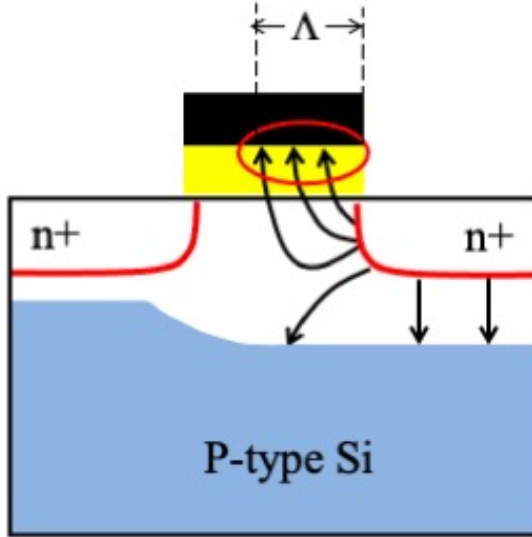


Fig. 3.10 Illustration of geometric screening in a bulk (or planar) MOSFET. The arrows represent the electric field lines caused by the positive voltage on the drain. Some field lines terminate on the metal gate and some in the p-type bulk. In a well-designed MOSFET, the distance that they penetrate, Λ , is less than L so that they do not reach through to the source and cause DIBL. (After David Frank, Yuan Taur, and Hon-Sum Philip Wong, “Future Prospects for Si CMOS Technology,” Technical Digest, IEEE Device Research Conf., pp. 18-21, 1999.)

From the demonstration of the first Si MOSFET in 1959 until the early 2000s, integrated circuit technology was largely bulk, planar, MOSFET technology. To preserve good electrical characteristics as L was scaled down, the gate oxide was thinned from 120 nm initially to just over 1 nm eventually. This put the gate electrode closer to the channel and increased its control of the potential at the top of the barrier. The channel doping was also increased to make it harder for the drain electric field to reach the source. Sophisticated channel doping profiles were developed (e.g. so-called halo implants), but to preserve good electrical characteristics at $L = 22$ nm and below, a new transistor structure was needed.

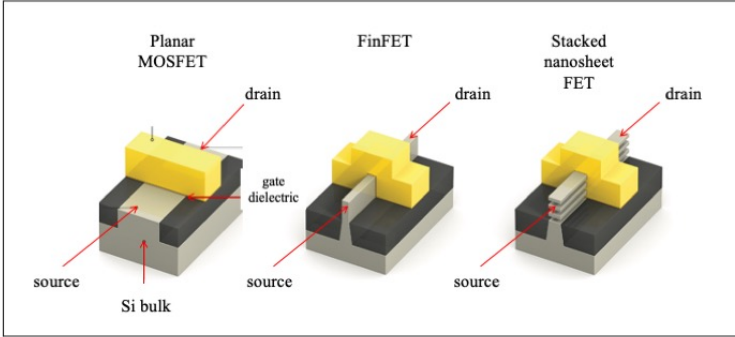


Fig. 3.11 Evolution of the MOSFET. Left: Bulk (planar) MOSFET. Middle: FinFET. Right: Stacked nanosheet MOSFET. (©IEEE 2019. Reprinted, with permission, from: Peide Ye, Thomas Ernst, and Mukesh V. Khare, “The Last Silicon Transistor” *IEEE Spectrum*, pp. 30-35, August, 2019.)

Figure 3.11 illustrates the evolution of the MOSFET. Beginning in 1959 with the demonstration of the first Si MOSFET and for the next 50 years, most MOSFETs had a planar geometry. Beginning in 2011 with the 22 nm technology node, adequate electrostatic control of short channel effects could not be achieved with the planar geometry. At that point, the industry shifted to the FinFET geometry shown in the middle of Fig. 3.11. In the FinFET, the channel is a vertical fin of silicon. Wrapping the gate around three sides of the fin gives the gate better control of top of the source to channel energy barrier. As channel lengths continue to scale down, eventually *gate all around* structures like the stacked nanosheet geometry shown at the right in Fig. 3.11 will be needed. In this structure, each Si nanosheet is encased in a thin gate dielectric covered by a metal gate electrode. By stacking nanosheets vertically, high on-currents can be achieved in a small transistor area. Such a device may be the ultimate field-effect transistor in terms of size and performance.

If the electrostatics can be controlled, is there a fundamental lower limit to the channel length? This question can be explored by numerical device simulations that treat electrons as quantum mechanical particles. Figure 3.12 shows simulations of quantum transport in a gate all around nanowire Si MOSFET. Electrostatic control for this model device is excellent, so the scaling limits are determined by quantum mechanical tunneling of electrons from the source through the source to channel barrier in the off-state. The plots show the energy-resolved current in the transistor.

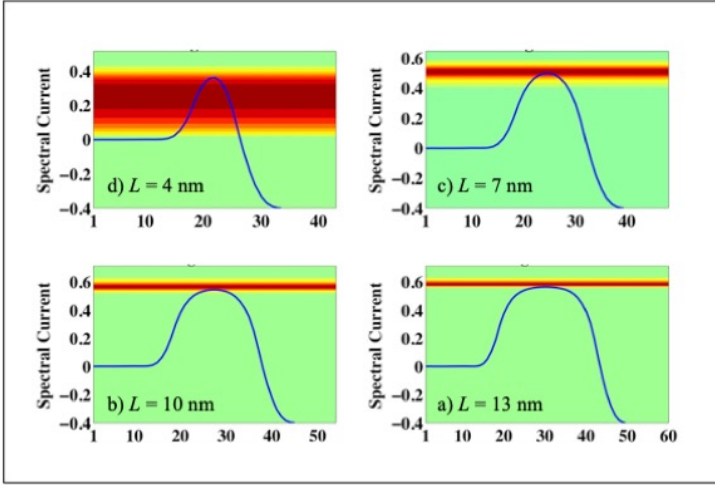


Fig. 3.12 The energy-resolved current as computed by a quantum transport simulation for a silicon nanowire MOSFET in the off-state under high drain bias. The nanowire is $\langle 110 \rangle$ oriented and is 3 nm in diameter. The x-axis is in arbitrary units. (Simulations performed by Dr. Mathieu Luisier, ETH Zurich and used with permission, 2014).

At $L = 13$ nm a) on the lower right, the off-state leakage current flows over the top of the barrier. This transistor is operating as a classical, barrier-controlled device. When the channel length decreases to 10 nm (b, lower left), a small fraction of the current begins to tunnel through the barrier. At 10 nm, however, good transistor performance is still obtained. For $L = 7$ nm (c, upper right), a substantial fraction of the off-state current flows by tunneling under the barrier. The performance of the device (e.g. its subthreshold slope) degrades. Finally, at $L = 4$ nm (d, upper left) most of the off-state current is due to tunneling through the barrier. At this channel length, it is difficult to modulate the current by controlling the barrier height because the barrier is transparent to electrons. When the barrier is too thin and becomes transparent to electrons, we no longer have a good transistor. Modern transistors are getting close to this fundamental limit.

3.10 Summary

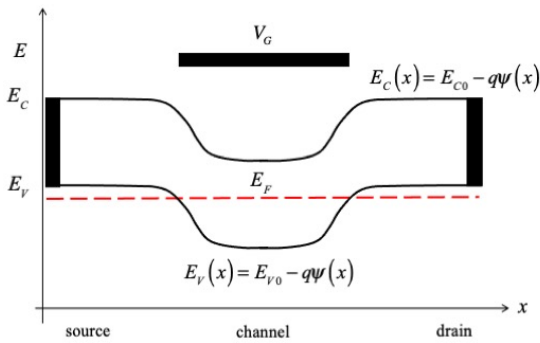
Transistor physics boils down to electrostatics and transport. The energy band diagram is a qualitative illustration of transistor electrostatics. In practice, most of transistor design is about engineering the device so that the energy barrier is appropriately manipulated by the applied voltages. The design challenges have increased as transistors have gotten smaller and smaller, and we understand transistor electrostatics better now, but the basic principles are the same as they were in the 1960s.

In a well-designed MOSFET, the region near the top of the barrier is under the strong control of the gate voltage and only weakly affected by the drain voltage. The goal in transistor design is to achieve this performance as channel length scaling brings the drain closer and closer to the source. The electrostatic design of MOSFETs has gotten more challenging as device dimensions have been scaled down over the past five decades, but the principles have not changed. The nature of electron transport in transistors has, however, changed considerably as transistors have become smaller and smaller. A proper treatment of transport in nanoscale transistors is essential to understanding these devices and will be our focus beginning in Lecture 7.

Finally, we note that the energy band diagrams that we have sketched are similar to the energy band diagrams for a bipolar transistor [1, 2]. In fact, the two devices both operate by controlling current by manipulating the height of an energy barrier [4]. The source of the MOSFET is analogous to the emitter of the BJT, the channel to the base of the BJT, and the drain to the collector of a BJT. This close similarity will prove useful in understanding the operation of short channel MOSFETs. In Lecture 8, we will briefly discuss bipolar transistors.

Lecture 3 Exercise: E-band diagrams for p-MOSFETs

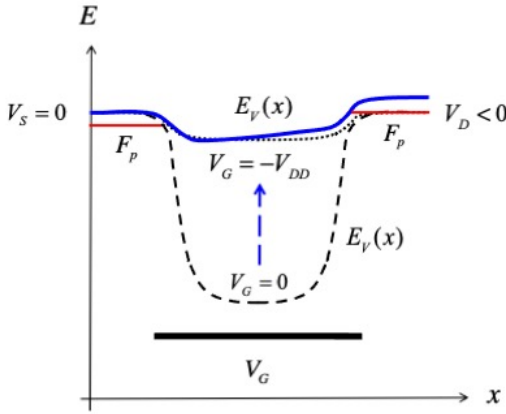
Our discussion of MOSFETs in terms of energy band diagrams, has been for n-channel MOSFETs. In this exercise, we'll examine the corresponding energy band diagrams for p-channel MOSFETs. Note that energy band diagrams are plots of *electron* energy vs. position. Electron energy increases upwards in these plots. We think about p-channel MOSFETs in terms of holes. The hole energy increases *downward* in these plots. The starting point for these plots is the equilibrium energy band diagram for a p-channel MOSFET shown below.



Equilibrium energy band diagram for a p-channel MOSFET. (Analogous to Fig. 3.3 for n-channel MOSFET.)

1) Sketch the energy band diagram for a PMOS transistor in the linear region.

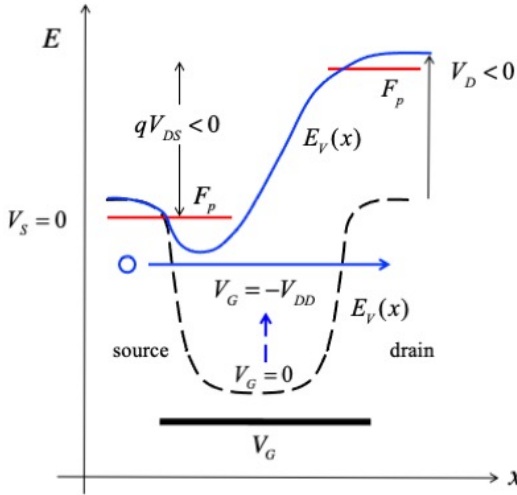
Just as we only plotted $E_c(x)$ for NMOS transistors, we will only plot $E_v(x)$ for a PMOS transistor. The transistor is on, so $V_G < V_{TP} < 0$, and the gate voltage raises the electron energy in the channel (i.e. lowers the hole energy). The drain voltage is negative, so it raises F_p and E_v in the drain. The electric field in the channel is positive, which pushes holes from the source to the drain. The energy band diagram for the linear region PMOS transistor is shown below.



Energy band diagram for a p-channel MOSFET biased in the linear region. The dashed line is $E_v(x)$ in equilibrium. (Analogous to Fig. 3.4b for an n-channel MOSFET.)

2) Sketch the energy band diagram for a PMOS transistor in the on-state.

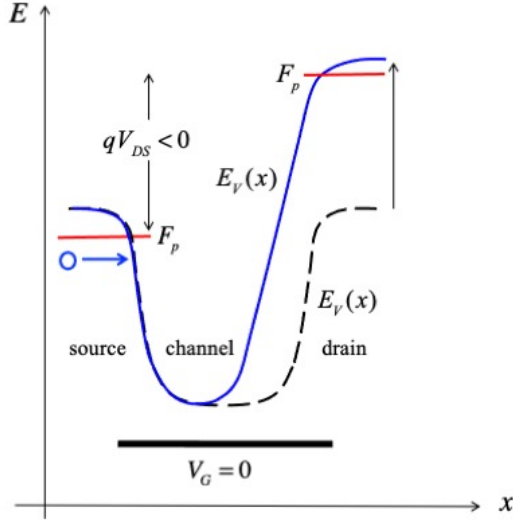
In this case, we are looking for the p-channel energy band diagram analogous to the on-state energy band diagram for the n-channel MOSFET, Fig. 3.5a. The result is shown below.



Energy band diagram for a p-channel MOSFET biased in the on-state. The dashed line is $E_v(x)$ in equilibrium. (Analogous to Fig. 3.5a for an n-channel MOSFET.)

3) Sketch the energy band diagram for a PMOS transistor in off-state.

We are looking here for the p-channel energy band diagram analogous to the off-state energy band diagram for the n-channel MOSFET, Fig. 3.5b. The result is shown below.



Energy band diagram for a p-channel MOSFET biased in the off-state. The dashed line is $E_v(x)$ in equilibrium. (Analogous to Fig. 3.5b for an n-channel MOSFET.)

3.11 References

Most of the important kinds of transistors are discussed in these texts:

- [1] Y. Taur and T. Ning, *Fundamentals of Modern VLSI Devices*, 2nd Ed., Oxford Univ. Press, New York, 2013.
- [2] Robert F. Pierret *Semiconductor Device Fundamentals*, 2nd Ed., Addison-Wesley Publishing Co., Reading, MA., 1996.

For a review of semiconductor fundamentals including the semiconductor equations and how current is related to gradients in the quasi-Fermi level, see:

- [3] Robert F. Pierret *Advanced Semiconductor Fundamentals*, 2nd Ed., Pearson Education, Inc., 2003.

Johnson describes the close relation of bipolar and field-effect transistors.

- [4] E.O. Johnson, “The IGFET: A Bipolar Transistor in Disguise,” *RCA Review*, **34**, pp. 80-94, 1973.